



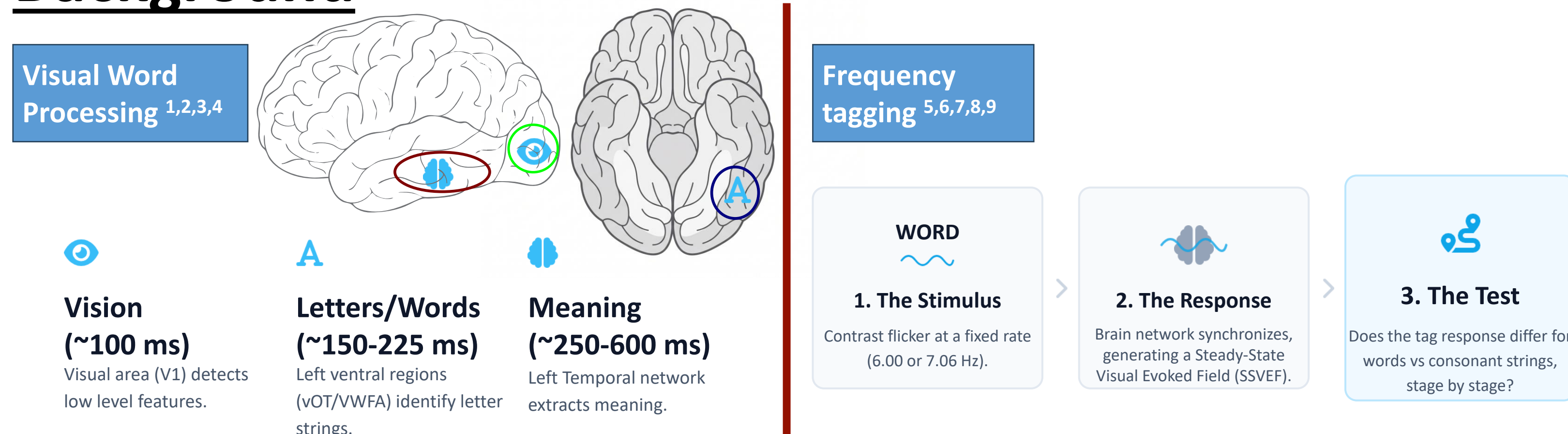
Characterizing word processing pathway using frequency tagging

Kishen Senziani¹, Berk Gercek¹, Nina Kazanina¹; ¹University of Geneva, Faculty of Medicine, Department of Basic Neurosciences

A 6–7 Hz flicker tag tracks the visual input, not wordness; the word-locked response (≥ 250 ms) separates words from consonant strings, maximally in anterior temporal cortex.

Update vs. submitted abstract: with fixation baselines and equal trial counts, the tag effect is null and the source effect is Word > Non-word.

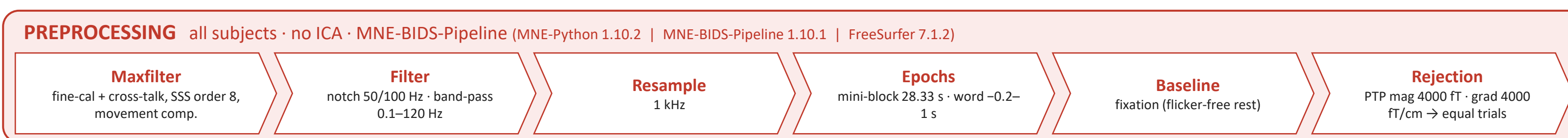
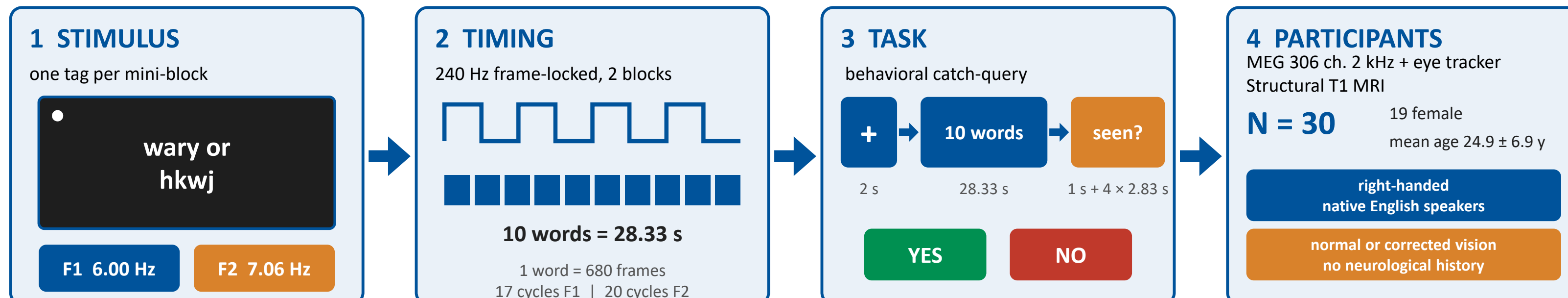
Background



AIM 1
Does the tag response (SSVEF) differ between words and consonant strings?

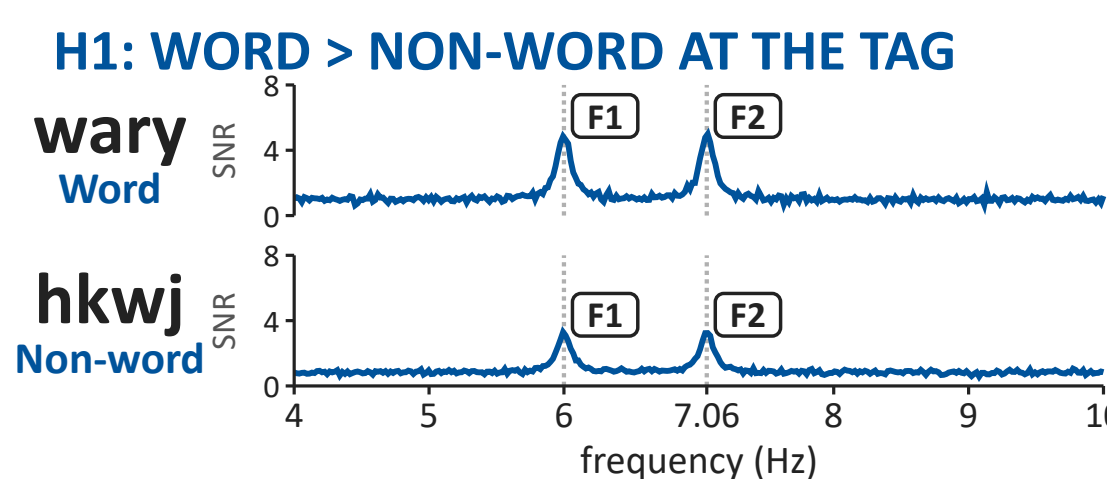
AIM 2
Does MEG source analysis recover the V1 $\rightarrow$ vOT $\rightarrow$ temporal sequence of the literature?

Methods



$$SNR(f) = \frac{\text{power at bin } f}{\text{mean of neighbour bins (13 or 3 per side)}}$$

Word vs Non-word
SNR contrast at the tag bin
paired t · TOST · cluster perm.



Pitfalls we hit — and how we fixed them

- Tags off by one FFT bin.** The display ran 0.6 % slow (photodiode: 5.965 / 7.024 Hz): read the SNR at the measured bin.
- Unequal trial counts faked an effect.** A source Word > Non-word effect vanished once counts were equalized.
- No gap between words.** The pre-stimulus window holds the previous word: baseline on fixation instead.
- Check the noise-covariance rank.** After SSS, 3 subjects kept a $\sim 1e\text{--}38$ eigenvalue ($dSPM \times 10^6$, distorted maps): fixed with `mne.compute_rank(tol='auto')`.
- Match the ink.** Consonant strings had ~ 10 % more ink and 20 % more height: the occipital tag followed it.



Code · supplementary figures S1–S4 · cohort
github.com/Orion212-lab/snl2026-ft-word-pathway

Cohort 30 recorded; each panel states its N. QC exclusions: no flicker on the photodiode, malformed blocks, no T1 MRI, all epochs rejected; sub-20/24 (source artifact); trigger anomalies sub-10–14 (main results unchanged without them). List via QR.

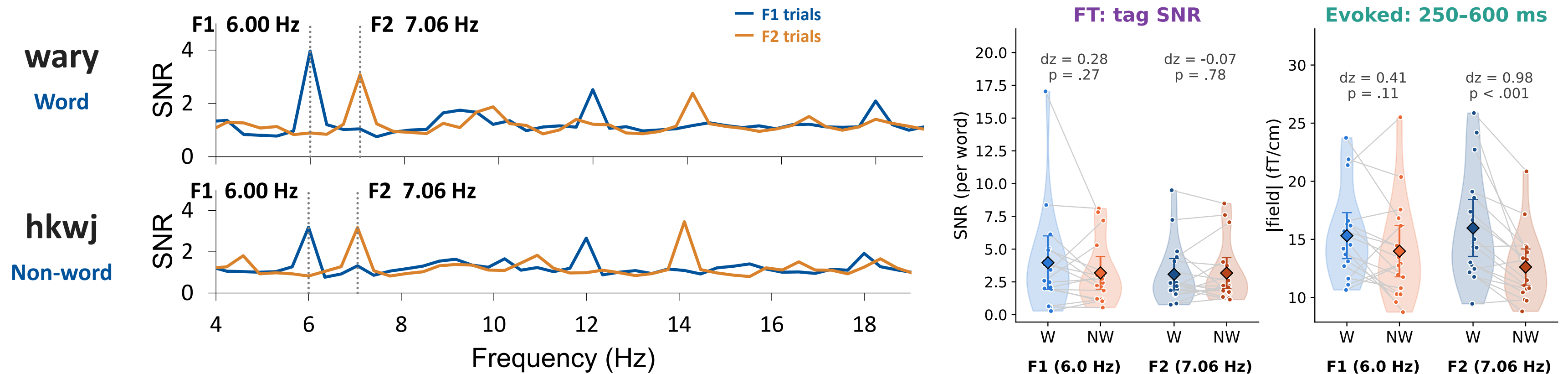
Confirmatory Word vs Non-word at the tag and in the evoked cascade windows (80–130, 150–225, 250–600 ms), left-temporal sensors (a priori). Everything else is exploratory.

Limits consonant strings also differ in orthography, pronounceability and ink · condition-pure blocks · late vOT effect fragile · catch-query accuracy not yet scored

Next pseudowords · item-rate (FPVS) tagging instead of contrast flicker · interleaved conditions · two-word phrase task (intermodulation), already recorded

Results

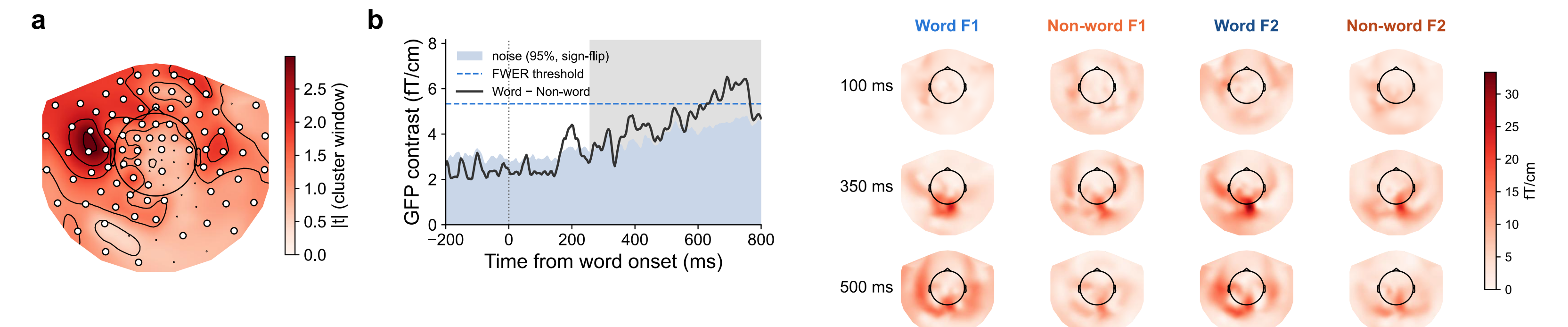
1 · The tag follows the visual input, not wordness (Aim 1)



Clear peaks at both tags and their harmonics. Left-temporal: Words = Non-words at either tag (F1 $dz = .28$, F2 $dz = -.07$; harmonics 1–4: equivalence $|dz| < .5$, TOST $p = .03$, BF01 = 4.2). Occipital, harmonics 1–4: Non-words > Words ($dz = -.88$, $p = .001$), matching their ~ 10 % more ink and 20 % greater height. The evoked response (250–600 ms) does separate Words at F2 ($dz = .98$, $p < .001$).

Left: per-word phase-locked SNR — epochs averaged per condition, one 2.83 s FFT ($df = 0.35$ Hz), 3 neighbour bins/side; left-temporal gradiometers. Right: same 17 subjects and per-word epochs for both DVs; W = Word, NW = Non-word; FT: tag SNR at the condition's own tag; evoked: mean $|field|$; diamonds = mean $\pm 95\%$ CI. Harmonics/occipital: N = 20.

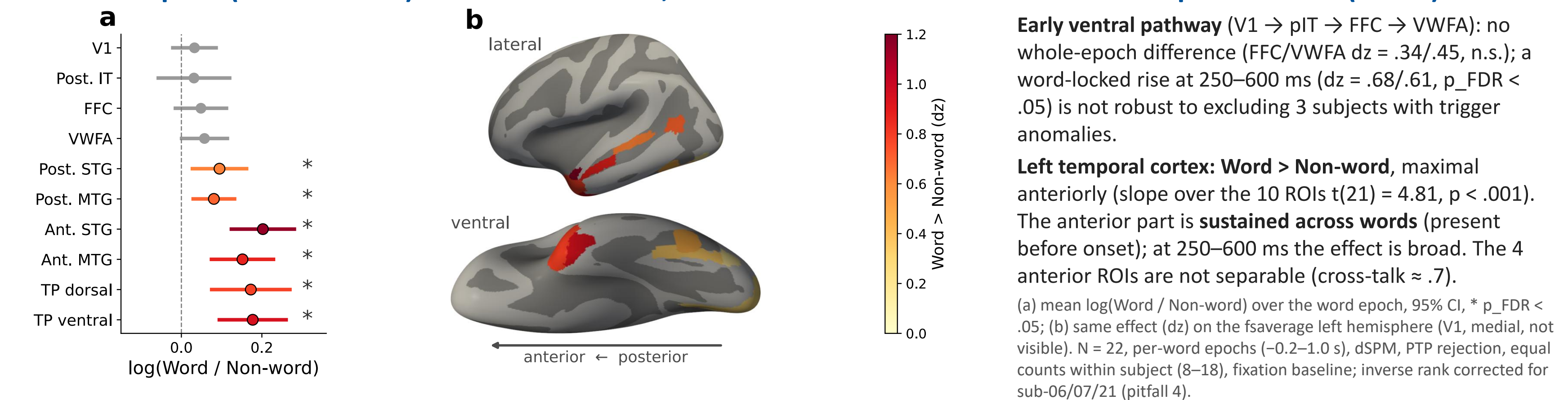
2 · Sensor space: the word-locked response separates words from consonant strings (Aim 2)



Later in the word epoch, Words exceed Non-words over left posterior and temporal sensors (gradiometers, cluster permutation, N = 26: one cluster, 84 sensors, $p < .001$; extent descriptive). Words drew more blinks ($dz = .94$), yet the effect holds in artifact-, blink- and saccade-free epochs ($dz = 1.31$, 20/21 subjects), without sub-20/24 and without 5 subjects with trigger anomalies. Tag $\times$ Wordness: n.s.

(a) mean $|t|$ in the cluster window (N = 26); (b) GFP of W – NW vs sign-flip noise (shaded: 95 %; dashed: FWER), exceeded at 628–764 ms; right: grand averages (N = 25), ± 25 ms; fixation baseline.

3 · Source space (individual T1): Word > Non-word, maximal and sustained in anterior temporal cortex (Aim 2)

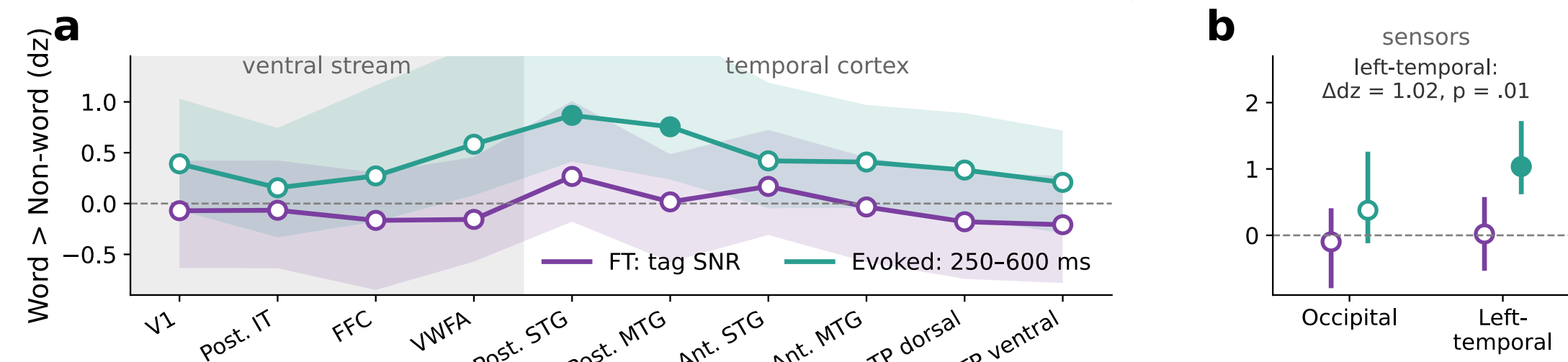


Early ventral pathway (V1 $\rightarrow$ pIT $\rightarrow$ FFC $\rightarrow$ VWFA): no whole-epoch difference (FFC/VWFA $dz = .34/.45$, n.s.); a word-locked rise at 250–600 ms ($dz = .68/.61$, $p_{FDR} < .05$) is not robust to excluding 3 subjects with trigger anomalies.

Left temporal cortex: Word > Non-word, maximal anteriorly (slope over the 10 ROIs $t(21) = 4.81$, $p < .001$). The anterior part is **sustained across words** (present before onset); at 250–600 ms the effect is broad. The 4 anterior ROIs are not separable (cross-talk = .7).

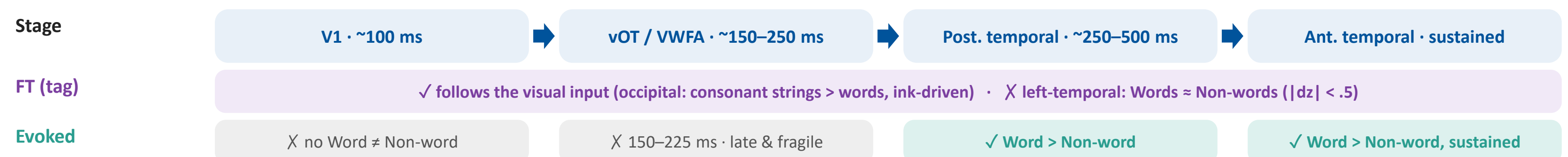
(a) mean $\log(\text{Word} / \text{Non-word})$ over the word epoch, 95% CI, * $p_{FDR} < .05$; (b) same effect (dz) on the fsaverage left hemisphere (V1, medial, not visible). N = 22, per-word epochs (–0.2–1.0 s), dSPM, PTP rejection, equal counts within subject (8–18), fixation baseline; inverse rank corrected for sub-06/07/21 (pitfall 4).

4 · Direct test: the word-locked response beats the tag, ROI by ROI (Aim 1)



Word-locked > tag, tested directly: the evoked effect (250–600 ms) lies above the tag effect in every ROI, significantly on the a-priori left-temporal sensors ($\Delta dz = 1.02$ [.29, 1.92], $p = .01$). The tag stays at zero along the pathway (large effects excluded, TOST $dz = .8$).

N = 17, same per-word epochs; FT: tag SNR, evoked: $|dSPM|$ 250–600 ms, both log. Bands/bars: bootstrap 95% CI; filled = $p < .05$ (FDR across the 10 source ROIs). 80–130 / 150–225 ms windows: QR.



Aim 1 — no: the tag indexes visual input, not wordness; the word-locked response does. **Aim 2 — partly:** temporal stages replicated, vOT only late and fragile, latency order n.s.